P1. $\varphi_{\alpha_i}(x)$ constitutes orthonormal set of states.

1) Equal quantum numbers $\alpha_1 = \alpha_2 = \alpha_3 = \alpha$ (only <u>bosons</u>)

$$\Rightarrow \psi = \varphi_\alpha(x_1)\,\varphi_\alpha(x_2)\,\varphi_\alpha(x_3)$$

this function is normalized, $\int dx_1 \int dx_2 \int dx_3\, |\psi(x_1, x_2, x_3)|^2 = 1$.

2) $\alpha_1 = \alpha_2 = \alpha$, $\alpha_3 \neq \alpha$ (only <u>bosons</u>)

$$\psi = c\Big(\varphi_\alpha(x_1)\varphi_\alpha(x_2)\varphi_{\alpha_3}(x_3) + \varphi_{\alpha_3}(x_1)\varphi_\alpha(x_2)\varphi_\alpha(x_3) + \varphi_\alpha(x_1)\varphi_{\alpha_3}(x_2)\varphi_\alpha(x_3)\Big)$$

To normalize, $\int dx_1 \int dx_2 \int dx_3\, |\psi(x_1, x_2, x_3)|^2 = 1$

$$\Rightarrow |c|^2 \int dx_1 dx_2 dx_3 \Big(\varphi_\alpha^*(x_1)\varphi_\alpha^*(x_2)\varphi_{\alpha_3}^*(x_3) + \varphi_{\alpha_3}^*(x_1)\varphi_\alpha^*(x_2)\varphi_\alpha^*(x_3) + \varphi_\alpha^*(x_1)\varphi_{\alpha_3}^*(x_2)\varphi_\alpha^*(x_3)\Big)\cdot\Big(\varphi_\alpha(x_1)\varphi_\alpha(x_2)\varphi_{\alpha_3}(x_3) + \varphi_{\alpha_3}(x_1)\varphi_\alpha(x_2)\varphi_\alpha(x_3) + \varphi_\alpha(x_1)\varphi_{\alpha_3}(x_2)\varphi_\alpha(x_3)\Big) = 1$$

since, for example, $\int dx_1\, \varphi_\alpha^*(x_1)\varphi_{\alpha_3}(x_1) = 0$,

there are only 3 non-zero terms $\Rightarrow c = \frac{1}{\sqrt{3}}$.

3) $\alpha_1 \neq \alpha_2 \neq \alpha_3 \Rightarrow$ fermions are possible now

$$\psi_F = \frac{1}{\sqrt{3!}} \begin{vmatrix} \varphi_{\alpha_1}(x_1) & \varphi_{\alpha_2}(x_1) & \varphi_{\alpha_3}(x_1) \\ \varphi_{\alpha_1}(x_2) & \varphi_{\alpha_2}(x_2) & \varphi_{\alpha_3}(x_2) \\ \varphi_{\alpha_1}(x_3) & \varphi_{\alpha_2}(x_3) & \varphi_{\alpha_3}(x_3) \end{vmatrix}$$

expand to get explicit form.

For bosons, replace all $\ominus$ signs in the expansion by $\oplus$ signs.

minus

P2. Arbitrary state in number-representation is written as

$$|\psi\rangle = \sum_{n=0}^{\infty} c_n |n\rangle$$

We are asked to find $|\psi\rangle$ such that

$$\hat{a}|\psi\rangle = \alpha|\psi\rangle$$

But $\hat{a}|\psi\rangle = \sum_{n=0}^{\infty} c_n \underbrace{\hat{a}|n\rangle}_{\sqrt{n}|n-1\rangle} = \sum_{n=0}^{\infty} c_n \sqrt{n}\,|n-1\rangle = \sum_{n=0}^{\infty} \alpha c_n |n\rangle$

$$\Rightarrow \cancel{c_0\sqrt{0}\,|-1\rangle} + c_1|0\rangle + c_2\sqrt{2}\,|1\rangle + \ldots =$$

$$= \alpha\left(c_0|0\rangle + c_1|1\rangle + c_2|2\rangle + \ldots\right)$$

That is, $c_1 = \alpha c_0$

$$\sqrt{2}\,c_2 = \alpha c_1 = \alpha^2 c_0 \Rightarrow c_2 = \frac{\alpha^2}{\sqrt{2}} c_0$$

$$\sqrt{3}\,c_3 = \alpha c_2 = \frac{\alpha^3}{\sqrt{2}} c_0 \Rightarrow c_3 = \frac{\alpha^3}{\sqrt{2\cdot 3}} c_0$$

$$\vdots$$

$$c_n = \frac{\alpha^n}{\sqrt{n!}} c_0$$

$$|\psi\rangle = \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} c_0 |n\rangle = c_0 \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

Normalize: $\langle\psi|\psi\rangle = 1 \Rightarrow \sum_{n=0}^{\infty} |c_0|^2 \frac{(\alpha^*)^n(\alpha)^n}{n!} \langle n|n\rangle = 1$

$$\Rightarrow |c_0|^2 \sum_{n=0}^{\infty} \frac{(|\alpha|^2)^n}{n!} = |c_0|^2 e^{|\alpha|^2} = 1$$

$$\Rightarrow c_0 = e^{-|\alpha|^2/2}$$

$\Rightarrow$ $|\psi\rangle = e^{-|\alpha|^2/2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$ is normalized eigenstate of $\hat{a}$.

P3. N Free bosons in volume V.

(a) $|\Psi_0\rangle = \frac{(a_0^+)^N}{\sqrt{N!}} |0\rangle$ i.e. all bosons are in the state with $\vec{k}=0$.

(b) $\hat{n}(\vec{r}) = \hat{\Psi}^+(r)\hat{\Psi}(r) = \frac{1}{V}\sum_{\vec{k}_1,\vec{k}_2} e^{i\vec{r}(-\vec{k}_1+\vec{k}_2)} a^+_{k_1} a_{k_2}$

(c) $\bar{n} = \langle\Psi_0|\hat{n}(r)|\Psi_0\rangle =$

$$= \frac{1}{V}\sum_{k_1 k_2} e^{i\vec{r}(\vec{k}_2-\vec{k}_1)} \langle\Psi_0|\hat{a}^+_{k_1}\hat{a}_{k_2}|\Psi_0\rangle$$

but $\hat{a}_{k_2}|\Psi_0\rangle = 0$ for all $\vec{k}_2 \neq 0$

$$\hat{a}_{k_2=0}|\Psi_0\rangle = \sqrt{N}\,|\tilde{\Psi}_0\rangle$$

↳ ground state of $(N-1)$ boson

$\Rightarrow \langle\Psi_0|\hat{a}^+_{k_1}\hat{a}_{k_2}|\Psi_0\rangle = N\,\delta_{k_1,0}\,\delta_{k_2,0}$

$\Rightarrow \bar{n} = \frac{N}{V}$

Number of particles in sub-volume v is

$$\hat{N}_v = \int_v d^3\vec{r}\;\hat{n}(r)$$

$$\langle\Psi_0|\hat{N}_v|\Psi_0\rangle = \int_v d^3\vec{r}\,\langle\Psi_0|\hat{n}(r)|\Psi_0\rangle = v\cdot\frac{N}{V}$$

(d) Start with

$$\langle\Psi_0|\hat{N}_v^2|\Psi_0\rangle = \langle\Psi_0|\int_v d^3\vec{r}_1\int_v d^3\vec{r}_2\;\hat{n}(\vec{r}_1)\hat{n}(\vec{r}_2)|\Psi_0\rangle$$

$$= \frac{1}{V^2}\sum_{k_1k_2}\sum_{k_3k_4}\int_v d^3\vec{r}_1\int_v d^3\vec{r}_2\; e^{i\vec{r}_1(\vec{k}_2-\vec{k}_1)}\, e^{i\vec{r}_2(\vec{k}_4-\vec{k}_3)}.$$

- $\langle \Psi_0 | \hat{a}^+_{k_1} \hat{a}_{k_2} \hat{a}^+_{k_3} \hat{a}_{k_4} | \Psi_0 \rangle$

This matrix element is finite only if $\vec{k}_1 = \vec{k}_4 = 0$ (otherwise $\hat{a}_{k_4} | \Psi_0 \rangle = 0$, and the same for its hermitian $\langle \Psi_0 | \hat{a}^+_{k_1} = 0$ if $\vec{k}_1 \neq 0$)

$$\Rightarrow \langle \Psi_0 | \hat{a}^+_0 \hat{a}_{k_2} \hat{a}^+_{k_3} \hat{a}_0 | \Psi_0 \rangle = N \langle \tilde{\Psi}_0 | \hat{a}_{k_2} \hat{a}^+_{k_3} | \tilde{\Psi}_0 \rangle$$

where $|\tilde{\Psi}_0\rangle = \dfrac{\hat{a}_0}{\sqrt{N}} |\Psi_0\rangle$

Next, Clearly $\vec{k}_2 = \vec{k}_3 = \vec{k}$ and if $\vec{k} = 0$ then

$$\langle \Psi_0 | \hat{a}^+_0 \hat{a}_0 \hat{a}^+_0 \hat{a}_0 | \Psi_0 \rangle = N^2$$

If $\vec{k} \neq 0$, $\langle \Psi_0 | \hat{a}^+_0 \hat{a}_k \hat{a}^+_k \hat{a}_0 | \Psi_0 \rangle =$

$$= \langle \Psi_0 | \hat{a}_k \hat{a}^+_0 \hat{a}_0 \hat{a}^+_k | \Psi_0 \rangle = N$$

Thus $\langle \Psi_0 | \hat{N}^2_v | \Psi_0 \rangle = \dfrac{1}{V^2} \displaystyle\int_v d^3\vec{r}_1 \int_v d^3\vec{r}_2 \left(N^2 + N \sum_{k \neq 0} e^{i\vec{k}(\vec{r}_1 - \vec{r}_2)} \right)$

We know that $\displaystyle\sum_{\text{all } \vec{k}} e^{i\vec{k}(\vec{r}_1 - \vec{r}_2)} = V \delta(\vec{r}_1 - \vec{r}_2)$

Hence $\displaystyle\sum_{k \neq 0} e^{i\vec{k}(\vec{r}_1 - \vec{r}_2)} = \sum_{\text{all } \vec{k}} e^{i\vec{k}(\vec{r}_1 - \vec{r}_2)} - \sum_{k=0} e^{i\vec{k}(\vec{r}_1 - \vec{r}_2)} =$

$$= V \delta(\vec{r}_1 - \vec{r}_2) - 1.$$

Then $\langle \Psi_0 | \hat{N}^2_v | \Psi_0 \rangle = \dfrac{1}{V^2} \displaystyle\int_v d\vec{r}_1 \int_v d\vec{r}_2 \left(N^2 + N V \delta(\vec{r}_1 - \vec{r}_2) - N \right)$

$$= \frac{N^2 v^2}{V^2} + \frac{N}{V} v - \frac{N v^2}{V^2} \Rightarrow \langle \Psi_0 | \hat{N}^2_v | \Psi_0 \rangle - \langle \Psi_0 | \hat{N}_v | \Psi_0 \rangle^2 =$$

$$= \frac{Nv}{V}\left(1 - \frac{v}{V}\right)$$